

MATH 123 LECTURE NOTES

WEEK ⑧

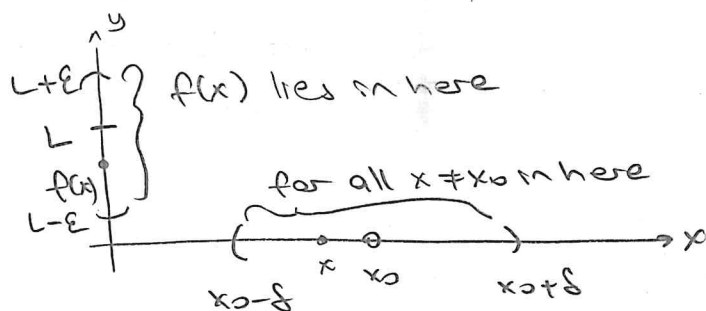
FORMAL DEFINITION OF LIMIT

Definition Let $f(x)$ be defined on an open interval about x_0 , except possibly at x_0 itself. We say that the limit of $f(x)$ as x approaches x_0 is the number L , and write

$$\lim_{x \rightarrow x_0} f(x) = L$$

If for every $\epsilon > 0$, there exists a corresponding $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \epsilon$$



Ex ① Show that $\lim_{x \rightarrow 1} 5x - 3 = 2$

Set $x_0 = 1$ $L = 2$. For any given $\epsilon > 0$, we have to find a suitable $\delta > 0$ so that if $x \neq 1$ and x is within distance δ of $x_0 = 1$, that is, whenever

$$0 < |x - 1| < \delta \text{ it is true } f(x) \text{ is within distance } \epsilon \text{ of } L = 2, \text{ i.e.}$$

$$|f(x) - 2| < \epsilon$$

$$|5x - 3 - 2| = |5x - 5| < \epsilon \quad \Rightarrow |x - 1| < \epsilon/5 \quad |x - 1| < \epsilon/5$$

Thus, we can take $\delta = \epsilon/5$. If $0 < |x - 1| < \delta = \epsilon/5$ then

$$|5x - 3 - 2| = |5x - 5| = 5|x - 1| < 5(\epsilon/5) = \epsilon$$

which proves $\lim_{x \rightarrow 1} 5x - 3 = 2$ (1)

IMPORTANT NOTE The value of $\delta = \epsilon/5$ is not the only value that will make $0 < |x-1| < \delta$ imply $|5x-5| < \epsilon$. Any smaller δ will do as well. The definition does not ask for a "best" positive δ , just one that will work.

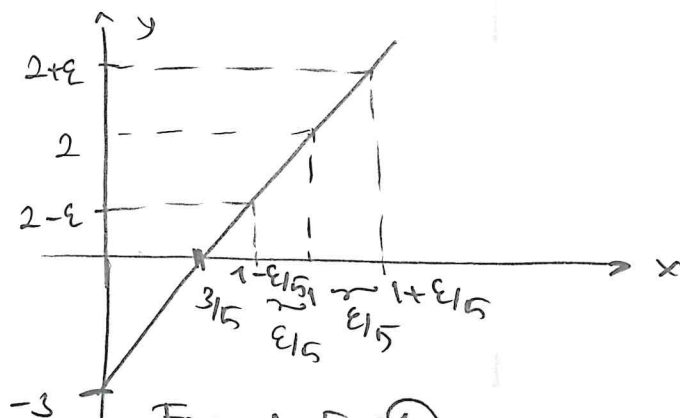


Fig of Ex 1

$0 < |x-1| < \epsilon/5$ guarantees that $|f(x)-2| < \epsilon$

Ex 2 $\lim_{x \rightarrow x_0} f(x) = y_0$. Show it?

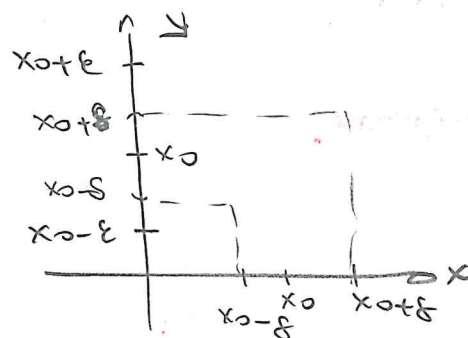


Fig of Ex 2

$0 < |x-x_0| < \delta$ will guarantee $|f(x)-y_0| < \epsilon$ whenever $\delta \leq \epsilon$.

Let $\epsilon > 0$ be given. We must find $\delta > 0$ such that for all x

$$0 < |x-x_0| < \delta \text{ implies } |f(x)-y_0| < \epsilon.$$

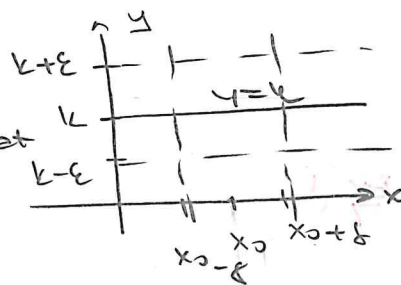
This will hold if $\delta = \epsilon$ or any smaller number.

Ex 3 $\lim_{x \rightarrow x_0} k = k$ ($k \in \mathbb{R}$)

Let $\epsilon > 0$ be given. We must find $\delta > 0$ such that for all x

$$|x-x_0| < \delta \text{ implies } |k-k| < \epsilon.$$

Since $k-k=0$, we can use any positive number for δ .



How to Find Algebraically δ for a given f, L, x_0 and $\epsilon > 0$

The process of finding a $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \epsilon$$

can be accomplished in two steps.

- ① Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (a, b) containing x_0 on which the inequality holds for all $x \neq x_0$.
- ② Find a value of $\delta > 0$ that places the open interval $(x_0 - \delta, x_0 + \delta)$ centered at x_0 inside the interval (a, b) . The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq x_0$ in this δ -interval.

Ex 7 For the limit $\lim_{x \rightarrow 5} \sqrt{x-1} = 2$, find $\delta > 0$ that works for $\epsilon = 1$.

① Solve the inequality $|\sqrt{x-1} - 2| < 1$

$$-1 < \sqrt{x-1} - 2 < 1 \Rightarrow 1 < \sqrt{x-1} < 3$$

$$1 < x-1 < 9 \quad 2 < x < 10$$

② Find a $\delta > 0$ for the interval $|x-5| < \delta$

$$5-\delta < x < 5+\delta$$

since we have $2 < x < 10$

we must take $\delta = 3$

Ex 7 For the limit $\lim_{x \rightarrow 0} \sqrt{x+1} = 1$ find $\delta > 0$ for $\epsilon = 0.1$.

$$|\sqrt{x+1} - 1| < 0.1$$

$$|x-0| < \delta$$

$$\delta = ?$$

Not involved

$$-0.1 < \sqrt{x+1} - 1 < 0.1$$

$$(0.9)^2 - 1 < x < (1.1)^2 - 1 \quad -0.21 < x < 0.21$$

$$0.9 < \sqrt{x+1} < 1.1$$

$$0.9^2 - 1 = -0.19 < x < 1.1^2 - 1 = 0.21$$

$$(0.9)^2 < x+1 < (1.1)^2$$

$$-\delta < x < \delta$$

Therefore, we

choose $\delta = 0.19$

③

Ex 7 Prove that $\lim_{x \rightarrow 2} f(x) = 4$ if $f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2. \end{cases}$

① Solve inequality

For $x \neq x_0 = 2$ we have $f(x) = x^2$ and the inequality to solve is $|x^2 - 4| < \epsilon$.

$$|x^2 - 4| < \epsilon \quad -\epsilon < x^2 - 4 < \epsilon$$

$$4 - \epsilon < x^2 < 4 + \epsilon \quad \sqrt{4 - \epsilon} < |x| < \sqrt{4 + \epsilon}$$

$$\sqrt{4 - \epsilon} < x < \sqrt{4 + \epsilon} \quad (\text{considers } \epsilon < 4)$$

② Find $\delta > 0$

Our aim is to find

$$|x - 2| < \delta \quad 2 - \delta < x < 2 + \delta$$

$$2 - \delta = \sqrt{4 - \epsilon} \Rightarrow \delta = 2 - \sqrt{4 - \epsilon}$$

$$2 + \delta = \sqrt{4 + \epsilon} \Rightarrow \delta = \sqrt{4 + \epsilon} - 2$$

$$\delta = \min \{ 2 - \sqrt{4 - \epsilon}, \sqrt{4 + \epsilon} - 2 \}$$

Ex 7 Prove $\lim_{x \rightarrow 3} (x^2 + 2x - 1) = 14$

$$|x^2 + 2x - 1| < \epsilon \Rightarrow |(x+5)(x-3)| < \epsilon$$

Our aim is to find $|x - 3| < \delta$

$$|x - 3| < \frac{\epsilon}{|x + 5|}$$

Let's choose δ arbitrary 1. This means that

$$|x - 3| < 1 \Rightarrow -1 < x - 3 < 1$$

$2 < x < 4$ This means

$$\text{that } 7 < x + 5 < 9 \Rightarrow |x + 5| < 9$$

$$|x + 5| \cdot |x - 3| < 9 \cdot |x - 3| < \epsilon$$

$$|x - 3| < \epsilon/9$$

$$\delta = \min \{ 1, \epsilon/9 \}$$

[If we choose $\delta = 1$ $|x - 3| < 1 < \epsilon/9$ $|x - 3| \cdot |x + 5| < \epsilon$
 If we choose $\delta = \epsilon/9$ it is obvious
 (4)]

Ex 7 Prove that

$$\lim_{x \rightarrow 2} \frac{1}{x} = 1/2.$$

① Solve inequality $\left| \frac{1}{x} - \frac{1}{2} \right| < \epsilon$ $\frac{|2-x|}{2|x|} < \epsilon$

We must put a bound for x , let's consider $|x-2| < 1$.

$$-1 < x-2 < 1 \quad 1 < x < 3 \quad \frac{1}{3} < \frac{1}{x} < 1 \Rightarrow \frac{1}{|x|} < 1$$

Then we get $\frac{|x-2|}{2|x|} < \frac{|x-2|}{2} < \epsilon$ $|x-2| < 2\epsilon$

$$\delta = \min(1, 2\epsilon)$$

Assume we choose $|x| < \epsilon$ and $|2-x| < 3|x| < 3\epsilon$

Let's consider $|x-2| < 3\epsilon$ This means that,

$$\frac{|x-2|}{x} < 3\epsilon$$

Not equal to ϵ therefore we didn't choose $|x|$ and we reverse it

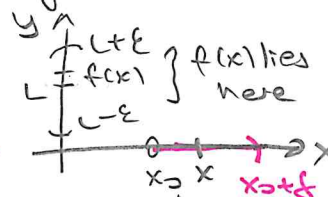
ONE-SIDED LIMITS

Definition We say that $f(x)$ has right limit L at x_0 , and write

$$\lim_{x \rightarrow x_0^+} f(x) = L$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$x_0 < x < x_0 + \delta \Rightarrow |f(x) - L| < \epsilon$$

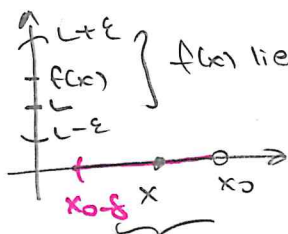


we say that $f(x)$ has left limit L at x_0 , and write for all $x \neq x_0$ in here

$$\lim_{x \rightarrow x_0^-} f(x) = L$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$x_0 - \delta < x < x_0 \Rightarrow |f(x) - L| < \epsilon$$



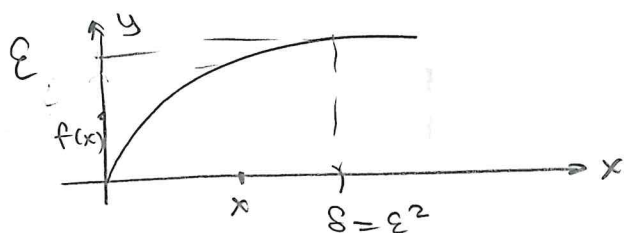
Ex 7 Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$

Let $\epsilon > 0$ be given. Here, $x_0 = 0$ and $L = 0$, so we want to find a $\delta > 0$ such that for all

$$0 < x < \delta \Rightarrow |\sqrt{x} - 0| < \epsilon$$

$$0 < x < \delta \Rightarrow \sqrt{x} < \epsilon \quad x < \epsilon^2$$

If we choose $\delta = \epsilon^2$ $0 < x < \delta = \epsilon^2 \Rightarrow \sqrt{x} < \epsilon$



DEFINITIONS

- ① We say that $f(x)$ has the limit L as x approaches infinity and write

$$\lim_{x \rightarrow \infty} f(x) = L$$

if for every number $\epsilon > 0$, there exists a corresponding number M such that for all x

$$x > M \Rightarrow |f(x) - L| < \epsilon$$

- ② We say that $f(x)$ has the limit L as x approaches minus infinity and write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if for every number $\epsilon > 0$, there exists a corresponding number N such that for all x

$$x < N \Rightarrow |f(x) - L| < \epsilon$$

Ex 7 Show that

① $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

② $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$

- ① We must find $x > M$ for $|\frac{1}{x} - 0| < \epsilon$

Let ϵ be given $|\frac{1}{x}| < \epsilon$ $\frac{1}{\epsilon} < |x|$ we can choose $M = 1/\epsilon$ this proves $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

- ② Let $\epsilon > 0$, we must find a number N such that

$$x < N \Rightarrow |\frac{1}{x} - 0| = |\frac{1}{x}| < \epsilon$$

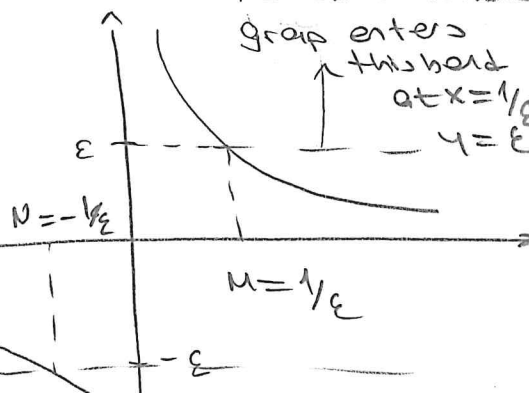
$$|x| > 1/\epsilon \text{ means}$$

This means that $N = -1/\epsilon$

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$$

no matter what positive number ϵ is the graph enters this band at $x = -1/\epsilon$ and stays

$x > 1/\epsilon$ or $x < -1/\epsilon$
For all ϵ values
graph enters this band at $x = 1/\epsilon$
 $y = \epsilon$



⑦

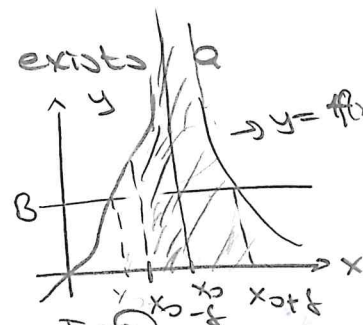
DEFINITIONS

- ① We say that $f(x)$ approaches infinity as x approaches x_0 , and write

$$\lim_{x \rightarrow x_0} f(x) = \infty$$

If for every positive real number B there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \Rightarrow f(x) > B$$

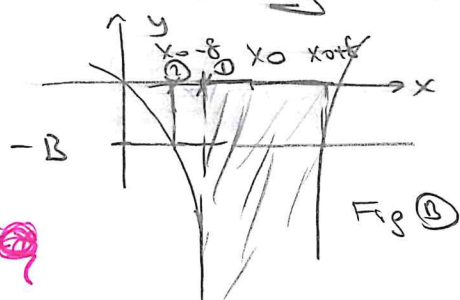


- ② We say that $f(x)$ approaches minus infinity as x approaches x_0 , and write

$$\lim_{x \rightarrow x_0} f(x) = -\infty$$

If for every negative real number $-B$, there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \Rightarrow f(x) < -B$$



($x_0 - \delta$ can be point ① or ② similarly $x_0 + \delta$ can be more left, the important thing is to satisfy inequality $|x - x_0| < \delta$ if $f(x) < -B$! (Similar is valid also for Fig A.

Ex 7 Prove that $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

Given $B > 0$, we want to find $\delta > 0$ such that

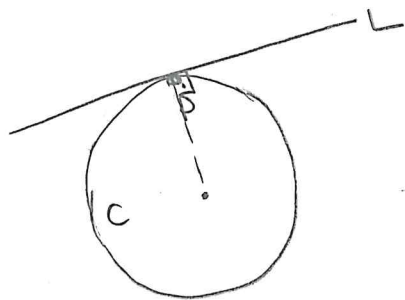
$$0 < |x - 0| < \delta \text{ implies } \frac{1}{x^2} > B$$

Now, $\frac{1}{x^2} > B$ if and only if $x^2 < \frac{1}{B}$

$$|x| < \frac{1}{\sqrt{B}} = \delta$$

DIFFERENTIATION

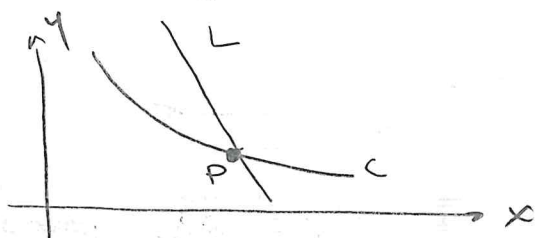
TANGENT LINES AND THEIR SLOPES



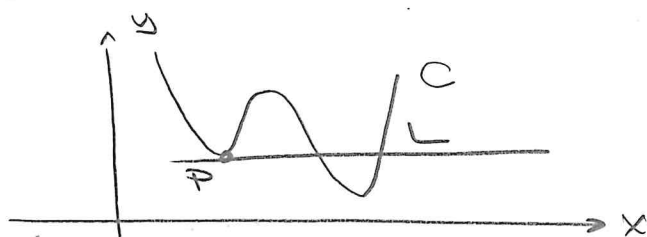
A tangent line (L) to the circle (C) has the following properties

- (i) It meets the circle at only one point
- (ii) The circle lies on only one side of the line
- (iii) The tangent is perpendicular to the line joining the center of the circle to the point of contact.

Let's check if these properties work for general definition of a tangent line of a curve.



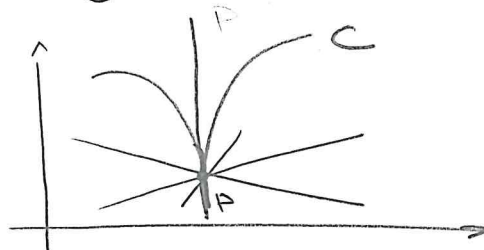
Not tangent (i) collapsed
meets at one point, but not tangent



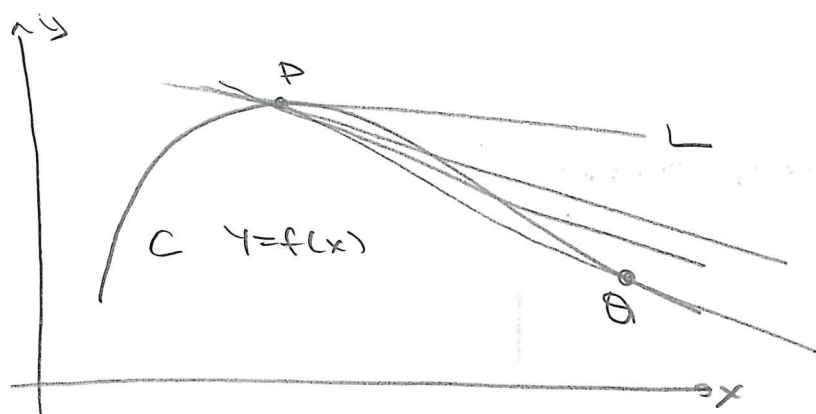
Tangent, but meets more than one points (ii collapsed);
C lies in two side of line (iii collapsed)



Tangent, but C crosses L (ii collapsed)



Many lines meet C only at P but none of them is tangent to C at P.



$$m_L = \lim_{Q \rightarrow P} m_{PQ}$$

Secant lines L passing through two different points of the curve C

PQ approach tangent line L as Q approaches P along the curve C

Definition Non vertical Tangent Lines

Suppose that the function f is continuous at $x=x_0$ and that

$$\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = m$$

exists. Then, the straight line having slope m and passing through the point $P=(x_0, f(x_0))$ is called the tangent line (or simply the tangent) to the graph of $y=f(x)$ at P .

An equation of this tangent line is

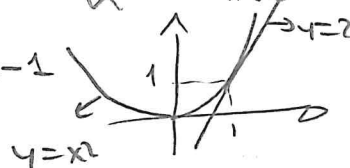
$$y = m(x - x_0) + y_0$$

Ex 7 Find an equation of the tangent line to the curve $y=x^2$ at the point $(1,1)$

$$m = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} = \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \rightarrow 0} \frac{2h + h^2}{h} = \lim_{h \rightarrow 0} (2 + h) = 2$$

Tangent line $y - 1 = 2(x - 1)$

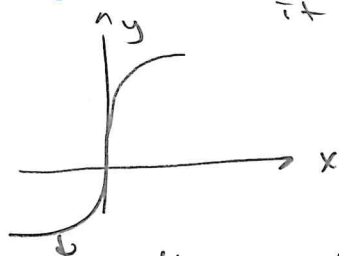
$$y = 2x - 1$$



Ex 7

Consider the graph of the function $y = x^{1/3}$ obtain its tangent line at point origin

$$y = x^{1/3}$$



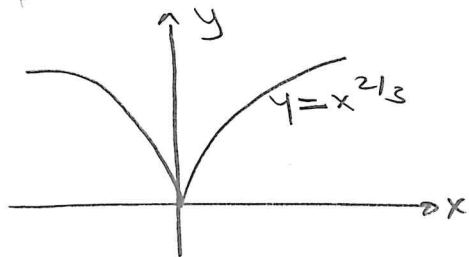
$$m = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^{1/3}}{h} = \lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = \infty$$

$m = \infty$ y -line ($x=0$ axis) is the tangent line

(Smooth curve)

(10)

Ex 7



$f(x) = x^{2/3}$ doesn't have a tangent at origin because it is not a smooth curve at origin

$$\frac{f(0+h) - f(0)}{h} = \frac{h^{2/3}}{h} = \frac{1}{h^{1/3}}$$

$$\lim_{h \rightarrow 0^+} \frac{1}{h^{1/3}} = \infty \quad \lim_{h \rightarrow 0^-} \frac{1}{h^{1/3}} = -\infty$$

We don't have limit at origin. We say that this curve has a cusp at the origin
(cusp = sharp, double, turning)

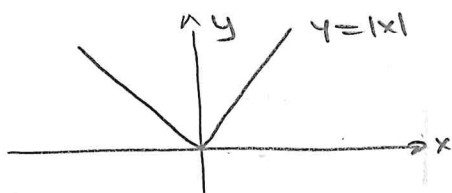
Definition Vertical Tangent Lines

If f is continuous at $P = (x_0, y_0)$ where $y_0 = f(x_0)$, and if either

$$\lim_{h \rightarrow 0} \underbrace{\frac{f(x_0+h) - f(x_0)}{h}}_{\text{Newton quotient (or a)}} = \infty \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = -\infty$$

then the vertical line $x = x_0$ is tangent to the graph $y = f(x)$ at P . If the limit Newton quotient fails to exist in any other way than being ∞ or $-\infty$ the graph $y = f(x)$ has no tangent line at P .

Ex 7 Does the graph of $y = |x|$ have a tangent line at $x = 0$?



$$\frac{|0+h| - |0|}{h} = \frac{|h|}{h} = \text{sgn}(h) = \begin{cases} +1 & h > 0 \\ -1 & h < 0 \end{cases}$$

Hence no limit, has no tangent line at $(0,0)$

Definition The slope of a Curve

The slope of a curve C at a point P is the slope of the tangent line C at P if such a tangent line exists.

In particular, the slope of the graph of $y = f(x)$ at the point P is

$$\lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$

Normals

If a curve C has a tangent line L at a point P , then the straight line N through P perpendicular to L is called the normal of C at P .

$$\text{Slope of the normal} = -\frac{1}{\text{Slope of the tangent}}$$

Ex 7 Find an equation of the normal to $y = x^2$ at $(1,1)$

$$m_T = 2 \quad m_N = -1/2 \quad \text{Normal line} \quad y - 1 = -1/2(x - 1)$$

Ex 7 Find equations of the straight lines that are tangent and normal to the curve $y = \sqrt{x}$ at the point $(4,2)$.

Sol

$$m_T = \lim_{h \rightarrow 0} \frac{\sqrt{4+h} - 2}{h} = \lim_{h \rightarrow 0} \frac{(\sqrt{4+h} - 2)(\sqrt{4+h} + 2)}{h(\sqrt{4+h} + 2)}$$
$$= \lim_{h \rightarrow 0} \frac{4+h-4}{h(\sqrt{4+h} + 2)} = \frac{1}{4}$$

$$m_T = \frac{1}{4} \quad \text{Tangent line} \quad y - 2 = \frac{1}{4}(x - 4)$$

$$m_N = -\frac{1}{m_T} = -4 \quad \text{Normal line} \quad y - 2 = -4(x - 4)$$

Ex 7 Find the tangent and normal line of $y = \frac{2x}{x+2}$ at $x=2$.

$$m_T = \lim_{h \rightarrow 0} \frac{f(h+2) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2(h+2)}{h+2+2} - 1}{h} = \lim_{h \rightarrow 0} \frac{\frac{2h+4}{h+4} - 1}{h}$$
$$= \lim_{h \rightarrow 0} \frac{\frac{2h+4 - h - 4}{(h+4) \cdot h}}{h} = \frac{1}{4}$$

For $x=2$, $y=1$

$$m_N = -4 \quad \text{Tangent} \quad y - 1 = \frac{1}{4}(x - 2)$$

$$\text{Normal} \quad y - 1 = -4(x - 2)$$

THE DERIVATIVE

Definition

The derivative of a function f is another function f' defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

at all points x for which the limit exists (i.e. is a finite real number). If $f'(x)$ exists, we say that f is differentiable at x .

Not**

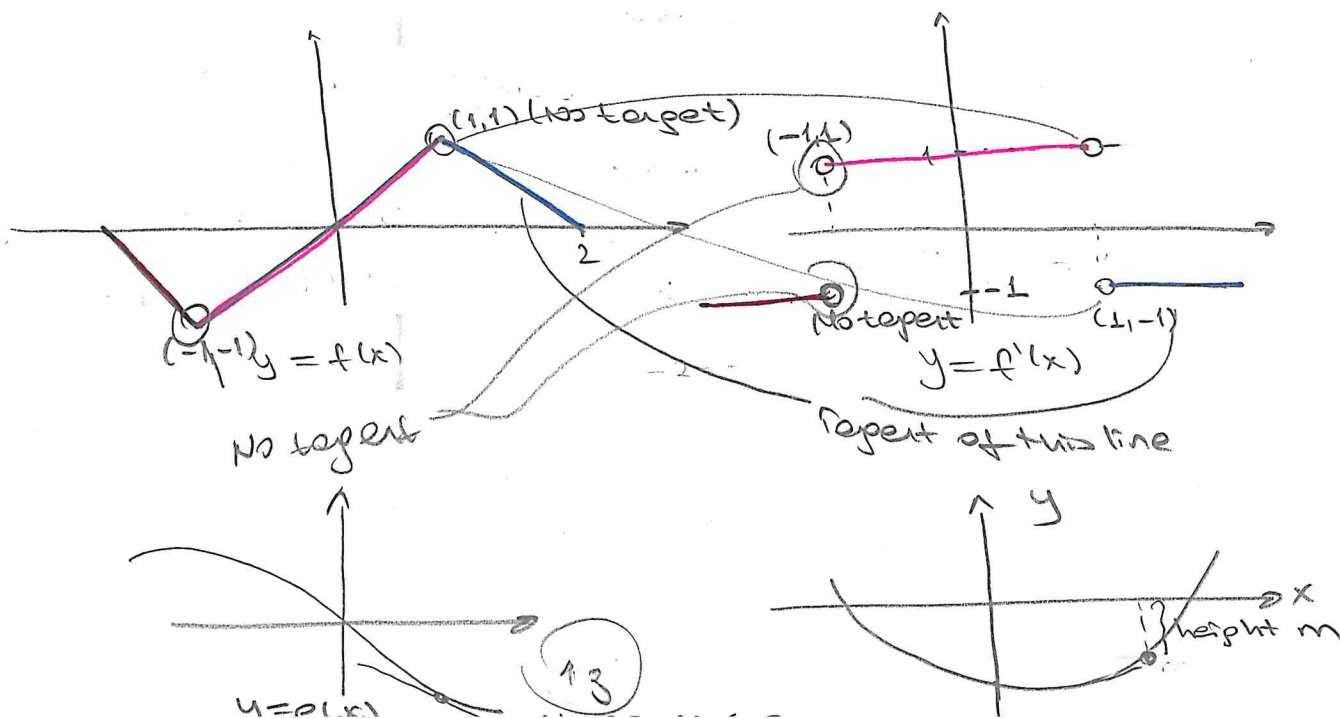
Values of $x \in D(f)$ where f is not differentiable and that are not endpoints of $D(f)$ are singular points of f .

REMARK

The value of the derivative of f at a particular point x_0 can be expressed as a limit in either of two ways.

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

In the second limit x_0+h is replaced by x , so that $h = x - x_0$ and $h \rightarrow 0$ is equivalent to $x \rightarrow x_0$.



Definition

$$f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \quad \text{right derivative at } x=a$$

$$f'_-(b) = \lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} \quad \text{left derivative at } x=b$$

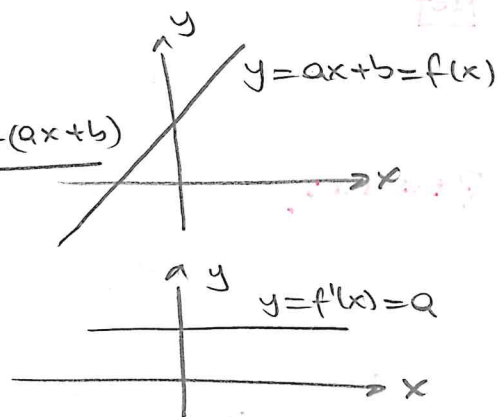
We say that f is differentiable on $[a,b]$ if $f'(x)$ exist for all x in (a,b) and $f'_+(a)$ and $f'_-(b)$ both exist.

Ex 7 (The derivative of a linear function) Show that if

$$f(x) = ax + b \quad \text{then } f'(x) = a$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a(x+h) + b - (ax + b)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{ah}{h} = a$$

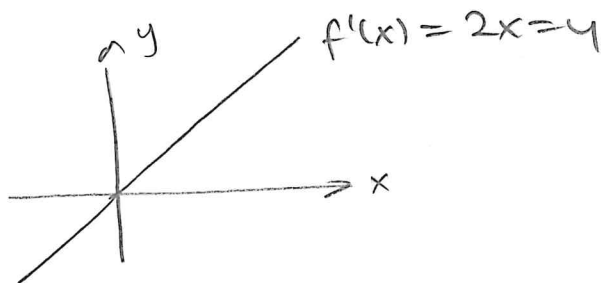
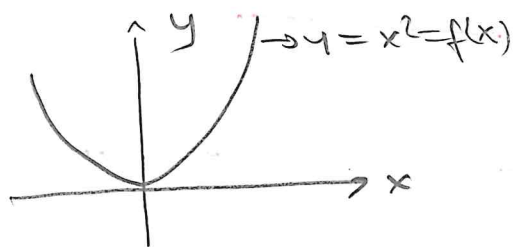


Ex 7 Use the definition of the derivative to calculate the derivatives of the functions

① $f(x) = x^2 \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

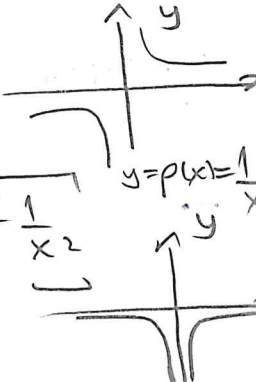
$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2hx + h^2 - \cancel{x^2}}{h} = \lim_{h \rightarrow 0} 2x + h = 2x$$



(b) $g(x) = \frac{1}{x}$ $g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$

$g'(x) = \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{h(x+h)x} = -\frac{1}{x^2}$

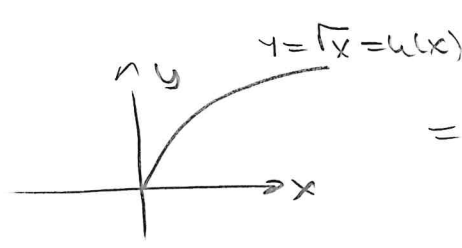


$y = g(x) = \frac{1}{x}$

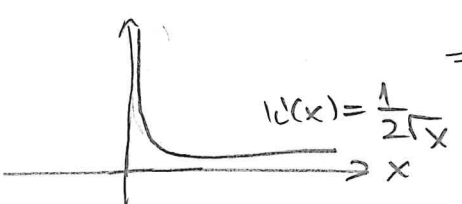
(c) $k(x) = \sqrt{x}$ $k'(x) = \lim_{h \rightarrow 0} \frac{k(x+h) - k(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$

$= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$

$= \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{2\sqrt{x}}$



$y = \sqrt{x} = k(x)$



$k'(x) = \frac{1}{2\sqrt{x}}$

General Power Rule:

If $f(x) = x^r$, then $f'(x) = r x^{r-1}$

This formula is valid for all values of r and x for each x^{r-1} makes sense as a real number.

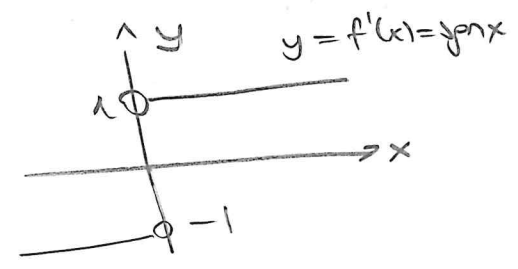
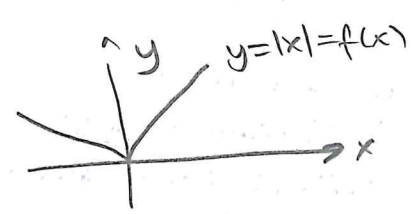
Ex If $f(x) = x^{5/3}$ $f'(x) = \frac{5}{3} x^{5/3-1} = \frac{5}{3} x^{2/3}$ for all real x

If $g(t) = \frac{1}{\sqrt{t}} = t^{-1/2}$ then $g'(t) = -\frac{1}{2} t^{-1/2-1} = -\frac{1}{2} t^{-3/2}$ for $t > 0$

Ex If $f(x) = |x|$, then $f'(x) = \frac{x}{|x|} = \text{sgn } x$

$f(x) = \begin{cases} x & x > 0 \\ -x & x < 0 \end{cases}$

$f'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$



Leibniz Notation

$y = f(x)$

$$D_x y = y' = \frac{dy}{dx} = \frac{d}{dx} f(x) = f'(x) = D_x f(x) = D f(x)$$

$$\frac{d}{dx} x^2 = 2x \quad \frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}} \quad \frac{d}{dt} t^{100} = 100t^{99}$$

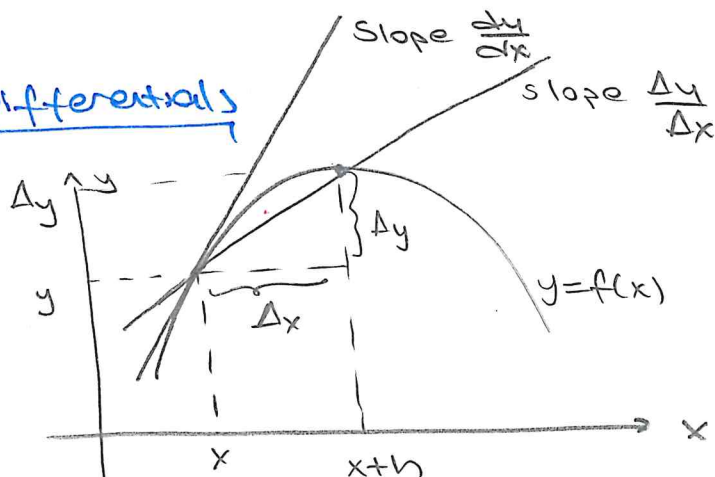
If $y = u^3$, then $\frac{dy}{du} = 3u^2$

The value of the derivative of a function at a particular number x_0 in its domain can be expressed in several ways:

$$D_x y \Big|_{x=x_0} = y' \Big|_{x=x_0} = \frac{dy}{dx} \Big|_{x=x_0} = \frac{d}{dx} f(x) \Big|_{x=x_0} = f'(x_0) = D_x f(x_0)$$

Ex $\frac{d}{dx} x^4 \Big|_{x=-1} = 4x^3 \Big|_{x=-1} = 4(-1)^3 = -4$

Differentials



Newton quotient.

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right)$$

Differential notation

regarding dx and dy as quantities (infinitesimal changes)

$$dy = \frac{dy}{dx} \cdot dx = f'(x) \cdot dx$$

as a new dependent variable (called the differential of y)

as a new independent variable (called the differential of x)

$$y = x^2 \Rightarrow dy = 2x \cdot dx \quad (\text{or } \frac{dy}{dx} = 2x)$$

$$f(x) = \frac{1}{x} \Rightarrow df(x) = -\frac{1}{x^2} dx \quad (16)$$

Thm Differentiability Implies Continuity

If f is differentiable at x , then it is continuous at x .

Proof Since f is differentiable at x we have

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

$$\text{Then } \lim_{h \rightarrow 0} (f(x+h) - f(x)) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) \cdot h = f'(x) \cdot 0 = 0$$

$$\lim_{h \rightarrow 0} f(x+h) = f(x) \text{ Continuous}$$

But the reverse is not correct: we CAN NOT say
(opposite)

~~If it is continuous it is differentiable~~

$$\text{Ex } f(x) = |x| \text{ is cont. at } x=0 \quad \left. \begin{array}{l} \lim_{x \rightarrow 0^+} x = 0 \\ \lim_{x \rightarrow 0^-} -x = 0 \end{array} \right\} \lim_{x \rightarrow 0} f(x) = f(0)$$

But not differentiable at $x=0$

$$\left. \begin{array}{l} f'_+(0) = 1 \\ f'_-(0) = -1 \end{array} \right\} \begin{array}{l} 1 \neq -1 \\ f'(0) \text{ not exist} \end{array}$$

DIFFERENTIATION RULES

THM If functions f and g are differentiable at x , and if C is a constant, then the functions $f+g$, $f-g$ and Cf are all differentiable at x and

$$(f+g)'(x) = f'(x) + g'(x)$$

$$(f-g)'(x) = f'(x) - g'(x)$$

$$(Cf)'(x) = C f'(x)$$

Proof

$$\begin{aligned}
 (f+g)'(x) &= \lim_{h \rightarrow 0} \frac{(f+g)(x+h) - (f+g)(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(f(x+h) + g(x+h)) - (f(x) + g(x))}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}
 \end{aligned}$$

$$(f+g)'(x) = f'(x) + g'(x)$$

Ex 7 Calculate the derivatives of the functions

(a) $f(x) = 2x^3 - 5x^2 + 4x + 7$

$f'(x) = 6x^2 - 10x + 4$

(b) $f(x) = 5\sqrt{x} + \frac{3}{x} - 18$

$f'(x) = \frac{5}{2\sqrt{x}} - \frac{3}{x^2}$

(c) $y = \frac{1}{7}t^4 - 3t^{7/3}$

$\frac{dy}{dt} = \frac{4}{7}t^3 - 3 \cdot \frac{7}{3} \cdot t^{4/3}$

Ex 7 Find an equation of the tangent to the curve $y = \frac{3x^2 - 4}{x}$ at the point on the curve where $x = -2$

$\frac{dy}{dx} = \frac{d}{dx} \left(3x^2 - \frac{4}{x} \right) = 6x + \frac{4}{x^2}$

$x = -2 \Rightarrow y = \frac{3(-2)^2 - 4}{-2} = -4$

$\left. \frac{dy}{dx} \right|_{x=-2} = -12 + \frac{4}{4} = -11$

Tangent line $y + 4 = -11(x + 2)$

Normal line $m_n = -\frac{1}{m_T} = \frac{1}{11}$

$y + 4 = \frac{1}{11}(x + 2)$

THM The Product Rule

If functions f and g are differentiable at x , then their product fg is also differentiable at x , and

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

Proof $(fg)'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \right)$$

$$= f'(x)g(x) + f(x)g'(x)$$

Ex Find the derivative of $(x^2+1)(x^3+4)$ using and without using the Product Rule

Product Rule $\frac{d}{dx}((x^2+1)(x^3+4)) = 2x(x^3+4) + 3x^2(x^2+1)$

Without product $(x^2+1)(x^3+4) = x^5 + 4x^2 + x^3 + 4$
 $\frac{d}{dx}(x^5 + 4x^2 + x^3 + 4) = 5x^4 + 8x + 3x^2$

Extension of product rule

$$(fgh)' = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$$

$$(f_1 f_2 f_3 \dots f_n)' = f_1' f_2 f_3 \dots f_n + f_1 f_2' f_3 \dots f_n + \dots + f_1 f_2 f_3 \dots f_n'$$

THM THE RECIPROCAL RULE

reciprocal (same as test)

If f is differentiable at x and $f(x) \neq 0$, then $1/f$ is differentiable at x , and

$$\left(\frac{1}{f}\right)' = -\frac{f'(x)}{(f(x))^2}$$

Proof

$$\begin{aligned}\frac{d}{dx} \left(\frac{1}{f(x)} \right) &= \lim_{h \rightarrow 0} \frac{\frac{1}{f(x+h)} - \frac{1}{f(x)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{f(x) - f(x+h)}{f(x)f(x+h)}}{h} \\&= \lim_{h \rightarrow 0} -\frac{1}{f(x+h) \cdot f(x)} \cdot \frac{f(x+h) - f(x)}{h} = -\frac{1}{(f(x))^2} f'(x)\end{aligned}$$

Ex Differentiate the functions

① $\frac{1}{x^2+1} = f(x)$ ② $f(t) = \frac{1}{t + \frac{1}{t}}$

③ $f'(x) = \frac{-2x}{(1+x^2)^2}$ ④ $f'(t) = \frac{(1 - \frac{1}{t^2})}{(1 + \frac{1}{t})^2}$

THM THE QUOTIENT RULE

quotient: both

If f and g are differentiable at x , and if $g(x) \neq 0$, then the quotient f/g is differentiable at x and

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$$

Proof

$$\begin{aligned}\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) &= \frac{d}{dx} \left(f(x) \cdot \frac{1}{g(x)} \right) = f'(x) \cdot \frac{1}{g(x)} + f(x) \left(-\frac{g'(x)}{(g(x))^2} \right) \\&= \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}\end{aligned}$$

Ex Find the derivatives of

$$\textcircled{a} \quad y = \frac{1-x^2}{1+x^2} \quad \frac{dy}{dx} = \frac{-2x(1+x^2) - 2x(1-x^2)}{(1+x^2)^2} = \frac{-4x}{(1+x^2)^2}$$

$$\textcircled{b} \quad y = \frac{\sqrt{t}}{3-5t} \quad \frac{dy}{dt} = \frac{\frac{1}{2\sqrt{t}}(3-5t) + 5\sqrt{t}}{(3-5t)^2} = \frac{3-5t+10t}{2\sqrt{t}(3-5t)^2}$$

$$\textcircled{c} \quad f(\theta) = \frac{a+b\theta}{m+n\theta} \quad \frac{df}{d\theta} = \frac{b(m+n\theta) - n(a+b\theta)}{(m+n\theta)^2} = \frac{bm-an}{(m+n\theta)^2}$$

Ex Find equation of any lines that pass through the point $(-1,0)$ and are tangent to the curve $y = (x-1)/(x+1)$

$$\frac{dy}{dx} = \frac{(x+1) - (x-1)}{(x+1)^2} = \frac{2}{(x+1)^2} \quad (-1,0) \text{ is not on the line } y = (x-1)/(x+1)$$

This means that we are searching a line that passes through $P = (a, \frac{a-1}{a+1})$ and $\theta = (-1,0)$

$$\left. \frac{dy}{dx} \right|_{x=a} = \frac{2}{(a+1)^2} \quad \text{slope of tangent at point P}$$

$$\text{Slope between } \theta P = \frac{\frac{a-1}{a+1} - 0}{a+1} = \frac{a-1}{(a+1)^2}$$

$$\text{So, } \frac{a-1}{(a+1)^2} = \frac{2}{(a+1)^2} \Rightarrow a-1=2 \quad a=3$$

$$\text{Then, the slope of the tangent line } \frac{dy}{dx} = \frac{2}{(4)^2} = \frac{2}{16} = \frac{1}{8}$$

$$\text{Tangent equation } y-0 = \frac{1}{8}(x-(-1)) \Rightarrow 8y = x+1$$

THE CHAIN RULE

THM The Chain Rule

If $f(u)$ is differentiable at $u = p(x)$, and $p(x)$ is differentiable at x , then the composite function $f \circ p(x) = f(p(x))$ is differentiable at x , and

$$(f \circ p)'(x) = f'(p(x)) p'(x)$$

→ In terms of Leibnitz notation, if $y = f(u)$ where $u = p(x)$, then $y = f(p(x))$ and

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} \quad \text{where } \frac{dy}{du} \text{ is evaluated at } u = p(x)$$

Ex Find the derivative of (a) $y = \sqrt{x^2+1}$

$$x^2+1 = u \quad y = \sqrt{u} \quad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{2\sqrt{u}} \cdot u' = \frac{1}{2\sqrt{x^2+1}} \cdot 2x$$

$$(b) (7x-3)^{10} = y \quad u = 7x-3 \quad y = u^{10}$$

$$\frac{dy}{dx} = 10u^9 \cdot u' = 10(7x-3)^9 \cdot 7$$

$$(c) f(t) = |t^2-1| \quad f'(t) = \text{sgn}(t^2-1) \cdot 2t = 2t \cdot \frac{t^2-1}{|t^2-1|} = \begin{cases} 2t & \text{for } t < -1 \text{ or } t > 1 \\ -2t & \text{for } -1 < t < 1 \\ \text{undef} & t = \pm 1 \end{cases}$$

$$(d) y = \left(3x + \frac{1}{(2x+1)^3} \right)^{1/4} \quad y = u^{1/4}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{4} u^{-3/4} \left(3 - \frac{3 \cdot 2x}{(2x+1)^4} \right)$$

$$= \frac{1}{4 \left(3x + \frac{1}{(2x+1)^3} \right)^{3/4}} \left(3 - \frac{3 \cdot 2}{(2x+1)^4} \right)$$

Building The Chain Rule into Differentiation Formulas

If u is a differentiable function of x and $y = u^n$, then

$$\frac{d}{dx} u^n = \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = n \cdot u^{n-1} \frac{du}{dx}$$

Ex 7 $\frac{d}{dx} \left(\frac{1}{u} \right) = -\frac{1}{u^2} \cdot \frac{du}{dx}$ $\frac{d}{dx} u^r = r u^{r-1} \frac{du}{dx}$

$$\frac{d}{dx} (\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx} \quad \frac{d}{dx} |u| = \operatorname{sgn} u \frac{du}{dx} = \frac{u}{|u|} \frac{du}{dx}$$

Derivatives of Trigonometric Functions

THM

$$\frac{d}{dx} \sin x = \cos x$$

Proof

$$\frac{d}{dx} \sin x = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x \cdot \cos h + \cos x \cdot \sin h - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \cos x \cdot \sin h}{h}$$

$$= \lim_{h \rightarrow 0} \sin x \underbrace{\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}}_0 + \lim_{h \rightarrow 0} \cos x \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\sin h}{h}}_1$$

$$= -\cos x$$

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} \cdot \frac{(\cos h + 1)}{(\cos h + 1)} = \lim_{h \rightarrow 0} \frac{\cos^2 h - 1}{h(\cos h + 1)} = \lim_{h \rightarrow 0} \frac{-\sin^2 h}{h(\cos h + 1)}$$

$$= \lim_{h \rightarrow 0} - \left(\frac{\sin h}{h} \right) \left(\frac{\sin h}{(\cos h + 1)} \right) = 0.$$

THM

$$\frac{d}{dx} \cos x = -\sin x$$

$$\begin{aligned} \frac{d}{dx} \cos x &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} = \lim_{h \rightarrow 0} \frac{\cos x \cdot \cos h - \sin x \cdot \sin h - \cos x}{h} \\ &= \lim_{h \rightarrow 0} \cos x \left(\frac{\cos h - 1}{h} \right) - \sin x \cdot \frac{\sin h}{h} = -\sin x \end{aligned}$$

Ex 7 Evaluate the following derivatives

(a) $\sin(\pi x) + \cos(3x)$ (b) $x^2 \sin \sqrt{x}$ (c) $\frac{\cos x}{1 - \sin x}$

(a) $\frac{d}{dx} (\sin(\pi x) + \cos(3x)) = \cos(\pi x) \cdot \pi - \sin(3x) \cdot 3$

(b) $\frac{d}{dx} (x^2 \sin \sqrt{x}) = 2x \cdot \sin \sqrt{x} + x^2 \cdot \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$

(c) $\frac{d}{dx} \left(\frac{\cos x}{1 - \sin x} \right) = \frac{-\sin x (1 - \sin x) - (-\cos x) \cdot \cos x}{(1 - \sin x)^2}$

Ex 7 Use two different methods to obtain the derivative of $f(t) = \sin t \cdot \cos t$

(I) $\frac{d}{dt} f = \frac{d}{dt} (\sin t \cdot \cos t) = \cos t \cdot \cos t - \sin t \cdot \sin t = \cos^2 t - \sin^2 t = \cos 2t$

(II) $f(t) = \sin t \cdot \cos t = \frac{\sin 2t}{2}$

$$\frac{df}{dt} = \frac{\cos 2t \cdot 2}{2} = \cos 2t$$

The Derivatives of Other Trigonometric Functions

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cdot \cot x$$

Ex 7 Find the tangent and normal lines to the curve

$$y = \left(\tan \frac{\pi x}{4} \right) \text{ at the point } (1, 1)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = \left(\sec^2 \frac{\pi x}{4} \right) \cdot \frac{\pi}{4} \Big|_{x=1} = \frac{\pi}{4} \sec^2 \left(\frac{\pi}{4} \right) = \frac{\pi}{4} (\sqrt{2})^2 = \pi/2$$

Tangent line $y - 1 = \pi/2 (x - 1)$

Normal line $m_N = -\frac{1}{m_T} = -\frac{2}{\pi}$ $y - 1 = -\frac{2}{\pi} (x - 1)$

Higher-Order Derivatives

→ If the derivative $y' = f'(x)$ of a function $y = f(x)$ is itself differentiable at x , we can calculate its derivative, which we call the second derivative of f and denote by

$$y'' = f''(x)$$

$$y'' = f''(x) = \frac{d^2 y}{dx^2} = \frac{d}{dx} \frac{d}{dx} f(x) = \frac{d^2}{dx^2} f(x) = D_x^2 y = D_x^2 f(x)$$

the n th derivative of $y = f(x)$ is

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n} = \frac{d^n}{dx^n} f(x) = D_x^n y = D_x^n f(x)$$

Ex 7 The velocity of a moving object is the rate of change of the position of the object w.r.t. time; if the object moves along the x-axis and is at position $x = f(t)$ at time t , then its velocity at that time is

$$v = \frac{dx}{dt} = f'(t)$$

Acceleration of the object $a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = f''(t)$

Ex 7 If $f(x) = x^n$ (where n is a positive integer), then

$$f^{(k)}(x) = n(n-1)(n-2)\dots(n-(k-1))x^{n-k}$$

$$= \begin{cases} \frac{n!}{(n-k)!} x^{n-k} & \text{if } 0 \leq k \leq n \\ 0 & \text{if } k > n \end{cases}$$

where $n!$ (called n factorial) is defined by

$$0! = 1$$

$$1! = 0! \cdot 1 = 1$$

$$2! = 1! \cdot 2 = 1 \cdot 2 = 2$$

$$3! = 3 \cdot 2! = 6$$

$$n! = (n-1)! \cdot n = 1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot (n-1) \cdot n$$

Ex 7 Show that $y = A \cos(kt) + B \sin(kt)$ $A, B, k \in \mathbb{R}$ is a solution of the second-order differential equation of

$$\frac{d^2y}{dt^2} + k^2y = 0 \quad \Rightarrow \quad \frac{dy}{dt} = -A \sin(kt) \cdot k + B \cdot k \cos(kt)$$

$$\frac{d^2y}{dt^2} = -A k^2 \cos(kt) - B k^2 \sin(kt)$$

$$-A k^2 \cos(kt) - B k^2 \sin(kt) + k^2 (A \cos(kt) + B \sin(kt)) = 0 \quad \checkmark$$

Ex 7 Find n th derivative of $y = \frac{1}{(1+x)} = (1+x)^{-1}$

$$y' = -(1+x)^{-2}$$

$$y''' = -2 \cdot 3 (1+x)^{-4}$$

$$y'' = 2 (1+x)^{-3}$$

$$y^{IV} = -2 \cdot 3 \cdot (-4) (1+x)^{-5}$$

$$y^V = -2 \cdot 3 \cdot (-4) \cdot (-5) (1+x)^{-6}$$

$$y^{(n)} = (-1)^n n! (1+x)^{-(n+1)}$$

EXAMPLES

Ex Using the definition of derivative obtain the tangent and normal lines of the functions at given points

① $y = (x-1)^2 + 1$ (2, 2)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h-1)^2 + 1 - [(x-1)^2 + 1]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) + 1 + 1 - (x-1)^2 - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2hx + h^2 - \cancel{2x} - 2h + \cancel{1} - \cancel{x^2} + \cancel{2x} - \cancel{1}}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h - 2 = 2x - 2$$

$$f'(x) = 2x - 2$$

$$m_T = f'(2) = 4 - 2 = 2$$

$$m_N = -\frac{1}{m_T} = -\frac{1}{2}$$

Tangent $\Rightarrow (y-2) = 2(x-2)$

Normal $\Rightarrow y-2 = -\frac{1}{2}(x-2)$

② $y = \frac{1}{x^2}$ (-1, 1)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} = \lim_{h \rightarrow 0} \frac{\cancel{x^2} - [x^2 + 2hx + h^2]}{(x+h)^2 \cdot x^2 \cdot h}$$

$$= \lim_{h \rightarrow 0} \frac{-2x-h}{(x+h)^2 \cdot x^2} = -\frac{2}{x^3}$$

$$m_T = f'(-1) = 2 \quad m_N = -\frac{1}{m_T} = -\frac{1}{2}$$

Tangent line $y-1 = 2(x+1)$

Normal line $y-1 = -\frac{1}{2}(x+1)$

③ $y = 2\sqrt{x}$ (1,2)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} - 2\sqrt{x}}{h} = \lim_{h \rightarrow 0} \frac{2\sqrt{x+h} - 2\sqrt{x}}{h} \cdot \frac{(2\sqrt{x+h} + 2\sqrt{x})}{(2\sqrt{x+h} + 2\sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{4(x+h) - 4x}{h(2\sqrt{x+h} + 2\sqrt{x})} = \lim_{h \rightarrow 0} \frac{4h}{h(2\sqrt{x+h} + 2\sqrt{x})} = \frac{1}{\sqrt{x}}$$

$m_T = f'(1) = 1$ $m_N = -1$

Repeat line $y - 2 = 1(x - 1)$

Normal line $y - 2 = -1(x - 1)$

④ $y = \frac{1}{x^3}$ $(-2, -1/8)$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^3} - \frac{1}{x^3}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{x^3 - (x+h)^3}{x^3(x+h)^3}}{h} = \lim_{h \rightarrow 0} \frac{x^3 - [x^3 + 3x^2h + 3xh^2 + h^3]}{x^3(x+h)^3 \cdot h}$$

$$= \lim_{h \rightarrow 0} \frac{-(3x^2 + 3hx + h^2)}{(x+h)^3 \cdot x^3} = \frac{-3x^2}{x^6} = -\frac{3}{x^4}$$

$m_T = f'(-2) = -\frac{3}{16}$ Repeat line $y + \frac{1}{8} = -\frac{3}{16}(x + 2)$

$m_N = \frac{16}{3}$ Normal line $y + \frac{1}{8} = \frac{16}{3}(x + 2)$

$f(x) = \begin{cases} x & x > 0 \\ -1 & x \leq 0 \end{cases}$ is it differentiable?

No. Because it is not continuous. $\lim_{x \rightarrow 0^+} x = 0 \neq \lim_{x \rightarrow 0^-} -1 = -1$

Differentiable $\Rightarrow$ Cont. ($P \Rightarrow Q$)

Not Cont $\Rightarrow$ Not Dif ($\neg Q \Rightarrow \neg P$)

(12)

Ex 7 $f(x) = \begin{cases} x^2 - 2x + 2 & x < 1 \\ -x^2 + 2x & x > 1 \end{cases}$ $f'(1) = ?$

$$f'_+(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^+} \frac{-(1+h)^2 + 2(1+h) - 1}{h}$$

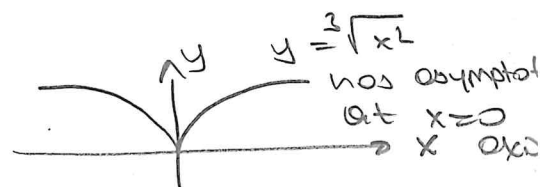
$$= \lim_{h \rightarrow 0^+} \frac{-1 - 2h - h^2 + 2h + 2 - 1}{h} = 0$$

$$f'_-(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0^-} \frac{(1+h)^2 - 2(1+h) + 2 - 1}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{1 + 2h + h^2 - 2h - 2 + 2 - 1}{h} = 0$$

$f'_+(1) = f'_-(1)$ so $f'(1) = 0$.

Ex 7 $f(x) = \sqrt[3]{x^2}$ $f'(0) = ?$



$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\sqrt[3]{h^2}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt[3]{h}} = \infty$$

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{1}{\sqrt[3]{h}} = -\infty$$

$f(x) = \sqrt[3]{x^2}$ doesn't have derivative at $x=0$.

Ex 7 Obtain the following derivatives.

① $y = 6x^2 - 10x - 5x^{-2}$

$$\frac{dy}{dx} = 12x - 10 + 10x^{-3}$$

② $y = \frac{12}{x} - \frac{4}{x^3} + \frac{1}{x^4}$

$$\frac{dy}{dx} = -\frac{12}{x^2} + \frac{12}{x^4} - \frac{4}{x^5}$$

③ $y = -2x^{-1} + \frac{4}{x^2} + \frac{x^2}{2}$

$$\frac{dy}{dx} = \frac{2}{x^2} - \frac{8}{x^3} + x$$

④ $y = (x^2+1)(x+\sqrt{x}+\frac{1}{x})$

$$\frac{dy}{dx} = 2x(x+\sqrt{x}+\frac{1}{x}) + (1-\frac{1}{x^2})(x^2+1)$$

$$\textcircled{5} \quad y = (1+x^2)(x^{3/4} - x^{-3})$$

$$\frac{dy}{dx} = 2x(x^{3/4} - x^{-3}) + \left(\frac{3}{4}x^{-1/4} + 3x^{-4}\right)(1+x^2)$$

$$\textcircled{6} \quad y = \frac{x^2-1}{x^2+x-2}$$

$$y' = \frac{2x(x^2+x-2) - (2x+1)(x^2-1)}{(x^2+x-2)^2}$$

Ex 7 If $f(2)=2$, $f'(2)=3$ obtain the following derivatives

$$\textcircled{1} \quad \frac{d}{dx} \left(\frac{x^2}{f(x)} \right) \Big|_{x=2} = \frac{2x \cdot f(x) - f'(x) \cdot x^2}{(f(x))^2} \Big|_{x=2} = \frac{4 \cdot f(2) - f'(2) \cdot 4}{(f(2))^2} = -1$$

$$\textcircled{2} \quad \frac{d}{dx} \left(\frac{f(x)}{x^2} \right) \Big|_{x=2} = \frac{f'(x) \cdot x^2 - 2x \cdot f(x)}{x^4} \Big|_{x=2} = \frac{f'(2) \cdot 4 - 4 \cdot f(2)}{16} = \frac{4}{16}$$

$$\begin{aligned} \textcircled{3} \quad \frac{d}{dx} \left(\frac{f(x)}{x^2 + f(x)} \right) \Big|_{x=2} &= \frac{f'(x)(x^2 + f(x)) - (2x + f'(x))f(x)}{(x^2 + f(x))^2} \Big|_{x=2} \\ &= \frac{f'(2)(4 + f(2)) - [4 + f'(2)]f(2)}{(4 + f(2))^2} = \frac{3[4+2] - [4+3] \cdot 2}{36} = \frac{4}{36} \end{aligned}$$

Ex 7 For the following functions obtain a, b values such that functions are differentiable everywhere.

$$\textcircled{1} \quad f(x) = \begin{cases} ax & x < 0 \\ x^2 - 3x & x \geq 0 \end{cases} \quad f'(x) = \begin{cases} a & x < 0 \\ 2x - 3 & x \geq 0 \end{cases}$$

$f'(x)$ is defined for $x < 0$ and $x > 0$. The critical integer is $x = 0$.

$$\begin{aligned} f'_+(0) &= 2 \cdot 0 - 3 = -3 \\ f'_-(0) &= a \end{aligned} \quad \text{To have } \underline{f'_+(0) = f'_-(0)} \quad \underline{a = -3}$$

$$(2) f(x) = \begin{cases} ax+b & x > -1 \\ bx^2-3 & x \leq -1 \end{cases}$$

$$f'(x) = \begin{cases} a & x > -1 \\ 2bx & x \leq -1 \end{cases}$$

for $x > -1$ and $x < -1$ it is defined. The critical integer

$$is \ x = -1$$

$$f'_+(-1) = a$$

$$\begin{array}{c} \xrightarrow{-1} \\ x < -1 \end{array} \in x > -1$$

$$\overline{a = -2b}$$

1. equation

$$f'_-(-1) = -2b$$

We need another equation. Since we have two unknowns a, b , so we need 2 equations. Therefore, we will use continuity

Differentiable $\Rightarrow$ Continuous.

$$\lim_{x \rightarrow -1^+} f(x) = -a+b$$

$$\lim_{x \rightarrow -1^-} bx^2-3 = b-3$$

$$\overline{b-3 = -a+b}$$

2. equation

$$a = 3$$

$$b = 3/2$$

Ex Obtain the following derivatives by using chain rule

$$(1) y = 6u - 9 \quad u = \frac{1}{2}x^4 \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 6 \cdot \frac{1}{2} \cdot 4x^3 = 12x^3$$

$$(2) y = \cos u \quad u = 3x+1 \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u \cdot 3 = \cos(3x+1) \cdot 3$$

$$(3) y = \cos u \quad u = \sin x \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = -\sin u \cdot \cos x = -\sin(\sin x) \cos x$$

$$(4) y = \tan u \quad u = 10x-5 \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (1+\tan^2 u) \cdot 10 = (1+\tan^2(10x-5)) \cdot 10$$

$$(5) y = \cos u \quad u = -\frac{x}{3} \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (-\sin u) \cdot (-\frac{1}{3}) = \sin(-\frac{x}{3}) \cdot \frac{1}{3}$$

$$(6) y = \sin u \quad u = x - \cos x \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos u (1 + \sin x)$$

$$(7) y = \left(1 - \frac{x}{7}\right)^{-7} \quad = \cos(x - \cos x) (1 + \sin x)$$

$$\frac{dy}{dx} = -7 \left(1 - \frac{x}{7}\right)^{-8} \cdot \left(-\frac{1}{7}\right)$$

$$\textcircled{8} \quad y = \left(\frac{x^2}{8} + x - \frac{1}{x} \right)^4$$

$$\frac{dy}{dx} = 4 \left(\frac{x^2}{8} + x - \frac{1}{x} \right) \left(\frac{x}{4} + 1 + \frac{1}{x^2} \right)$$

$$\textcircled{9} \quad y = \sec(\tan x) \Rightarrow y' = \sec(\tan x) \cdot \tan(\tan x) \cdot \sec^2 x$$

$$(\sec x)' = \sec x \cdot \tan x$$

$$(\tan x)' = \sec^2 x$$

$$\textcircled{10} \quad y = 5 \cos^{-4} x \quad y' = -20 \cos^{-5} x (-\sin x)$$

$$\textcircled{11} \quad r = (\cos \theta + \cot \theta)^{-1} \Rightarrow \frac{dr}{d\theta} = -(\cos \theta + \cot \theta)^{-2} \cdot (-\sin \theta - \csc^2 \theta)$$

$$\textcircled{12} \quad r = x^2 \sin^4 x + x \cos^{-2} x$$

$$\frac{dr}{dx} = 2x \cdot \sin^4 x + 4 \sin^3 x \cos x \cdot x^2 + \cos^{-2} x + x (-2) \cos^{-3} x (-\sin x)$$

$$\textcircled{13} \quad y = x \cdot \tan(2\sqrt{x}) + 7$$

$$\frac{dy}{dx} = \tan(2\sqrt{x}) + (1 + \tan^2(2\sqrt{x})) \cdot \frac{1}{\sqrt{x}} \cdot x$$